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JavaScript HTML DOM (Document Object Model)

20m

In this section, you will learn how you can use JavaScript to interact with the HTML DOM.

The DOM is a programming interface for HTML and extensible markup language (XML) documents. It defines the **logical structure** of documents and the way a document is accessed and manipulated.

The DOM is a way to represent the web page in a structured hierarchical way so that it will become easier for programmers and users (indirectly, though using the web content) to go through the document. Basically, the DOM is an API that represents and interacts with HTML or XML documents.



If you need a refresher about this, we recommend taking a look at the following pages:

[Document Object Model \(DOM\)](#) ➔

Why the DOM is used in JavaScript

HTML is used to **structure** web pages, and JavaScript is used to add **behaviour** to web pages. JavaScript cannot understand an HTML document directly when it is loaded into a browser; it needs a corresponding document to be created, which is the DOM.

The DOM is basically the representation of the same HTML document but in a different format with the use of objects.

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JavaScript can understand and interpret the DOM easily. For example, JavaScript cannot understand the heading tag This is a heading when it is within an HTML document; however, JavaScript can understand the object h1 in a DOM.

Because of the DOM, JavaScript can access each object (h1, p, body) by using different functions.

Why is it called an object model?

Documents are modelled using objects, and the model includes not only the structure of a document but also the behaviour of a document and the objects of which it is composed like tag elements with attributes in HTML.

Watch *The JavaScript DOM explained in 5 minutes!* (Bro Code, 2023) (5:00), to learn about the DOM. This video gives an overview of what the DOM is and how it works.

Select Play to begin.



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The JavaScript DOM explained in 5 minutes!

The JavaScript DOM explained in 5 minutes! 🌳

Bro Code



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All right. What's going on, people? So today I got to explain what the DOM is in JavaScript. The DOM is the Document Object Model. It's a JavaScript object that represents the page you see in the web browser, and it provides you with an API to interact with it. The web browser constructs the DOM when it loads an HTML document, and structures all of the elements in a tree-like representation.

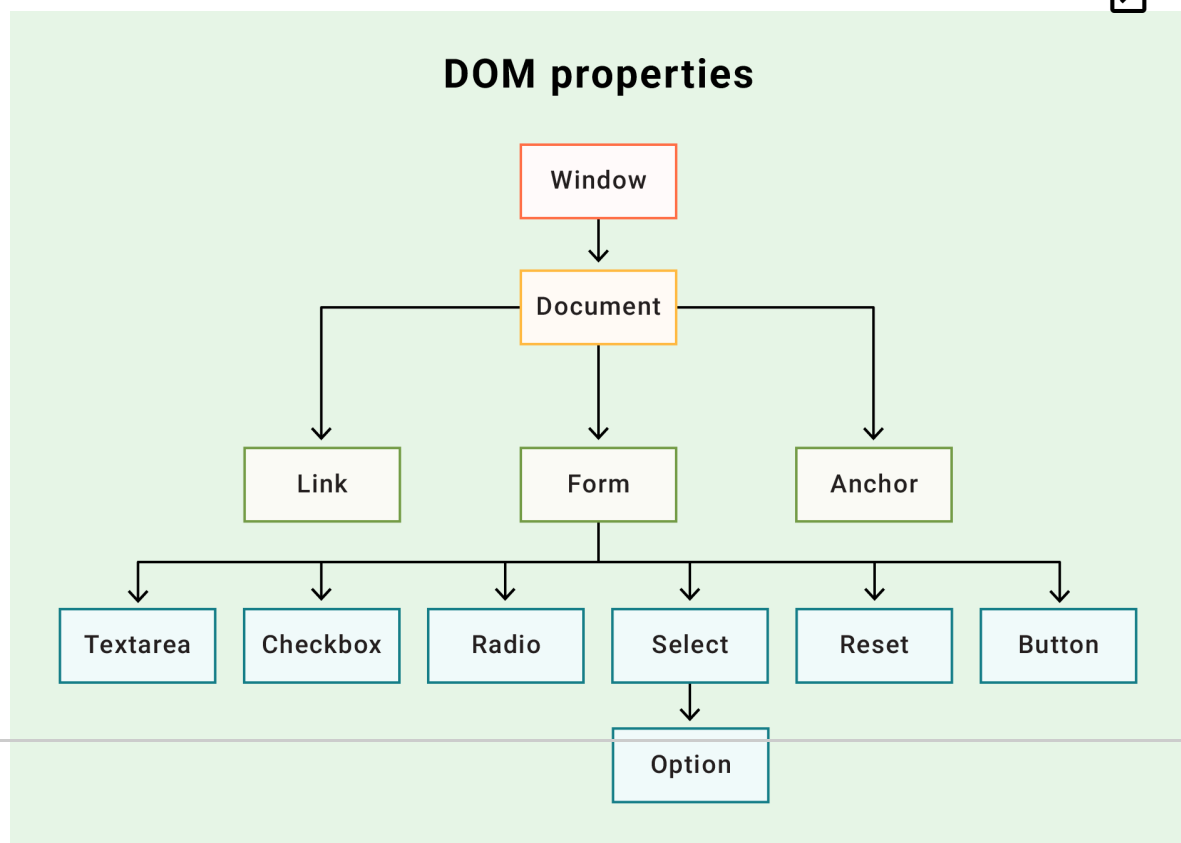
Within my HTML document, we have our HTML element as the root element. The HTML element contains a head element and a body element. And many various elements can be found within each of these. They're arranged in a tree-like data structure. But we access this tree-like data structure from the document object. In past lessons, in order to select an element by its ID, we would type document, dot, then follow it with a method, like getElementById. And then we would select an element by its ID.

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The figure below shows the high-level view of the properties of a DOM and how they are related. The properties of the document object can be accessed and modified by the document object. For example:



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Let's break down this DOM tree into its objects and properties.

Object	Description
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Window object	The window object is always at the top of the hierarchy of a DOM. It represents the browser window that will display the contents of the web page. It is similar to an API, which is used to set and access all the properties and methods of the browser. It is automatically created by the browser.
Document object	When an HTML document is loaded into a window, it becomes a document object. The document object represents the whole HTML document. It refers to other objects and properties that allow access to and modification of the content of the web page.
Form object	Form objects are represented by <form> tags. The form objects can contain many different form control elements. These are elements such as text fields, buttons, radio buttons or checkboxes.
Link object	Link objects are represented by <link> tags.
Anchor object	Anchor objects are represented by <href> tags.



Methods of document object

The HTML DOM methods present in JavaScript will perform actions on HTML elements.

For example, look at the following, where the HTML element is the 'mylabel'.

```
document.getElementById('mylabel');
```

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A method is an action you can do, such as adding or deleting an HTML element.

Method	Description	Syntax
<code>write('string')</code>	Is used to write a string into the document.	<code>document.write(exp1, exp2, ..., expN)</code>
<code>getElementById()</code>	It returns an element that matches the ID value given in the method.	<code>document.getElementById(id);</code>
<code>getElementsByName()</code>	It returns all the elements that match the name attribute given in the method.	<code>document.getElementsByName(nameAttribute);</code>
<code>getElementsByTagName()</code>	It returns all the elements that match the tag name given in the method.	<code>document.getElementsByTagName(tagName);</code>
<code>getElementsByClassName()</code>	It returns all the elements that match the class name given in the method.	<code>document.getElementsByClassName(className);</code>



HTML DOM properties

Each method of document object has a set of properties.

Each object that has been defined in the DOM has a set of methods and properties. We've already discussed the methods, so we'll now discuss the properties.

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A property is a value that you want to get or set for a particular HTML element. Common DOM properties are explained below. In the first example, we created an HTML element, p, then inserted some HTML into this element reference as *node*.

Property	Description	Example
.innerHTML	The innerHTML property sets or returns the <u>HTML content</u> of an element. It can access the nested <u>HTML</u> in an element and modify it.	<pre>const node = document.createElement('p '); node.innerHTML=' Forceful Hello! '</pre>
.textContent	The textContent property sets or returns the text content of a specified element. This property returns just the text content of an element and allows us to change that text.	<pre>node.textContent='Hello Again!'</pre>
.value	The value property sets or returns the value of an attribute. The value property can contain a default value or a value that the user types into a field.	<pre>node.value = ' Text changed in the text field '</pre>
.style.color	The color property sets or returns the colour of the text for an element.	<pre>node.style.color = 'red';</pre>



In Example 7.2, the JavaScript code shown below is used to display the results in a paragraph using the DOM.

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```
document.getElementById("result").innerHTML = "Addition : " + add;
```



HTMLJSResult

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LIVE

```
<!DOCTYPE html>
<html>

<head>
  <title>DOM manipulation</title>
</head>

<body>
  <label>Enter Value 1: </label>
  <input type="text" id="val1" />
  <br />
  <br />
  <label>Enter Value 2: </label>
  <input type="text" id="val2" />
  <br />
  <button onclick="getAdd()">Click
To Add</button>
  <p id="result"></p>
</body>
</html>
```

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Example 7.2 DOM method



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DOM

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Answer these questions about the code:

1. Identify the HTML element the JavaScript function has referenced.
2. Identify the part of the code that acts as the method (an action). Explain what this method does.
3. Identify the part of the code that acts as the property (the value). Explain what this property does.

1. result

2. getAdd() - The method takes Val1 and Val2 and combines them, the getAdd is updated to be a paragraph showing the output. The function also turns result to red

3.

29 / 50 Word Limit

1. <p id='result'></p> is the HTML element this code is referring to.
2. The method part of the code is document.getElementById('result'). It is finding the HTML element with the id attribute 'result'.
3. The property part of the code is .innerHTML. It is taking the value of the <p> element.

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Dig deeper

To explore more about the DOM and its methods and properties, read these resources:

- *Explain Various Type of HTML DOM Methods* (GeeksforGeeks, 2025), which explains a few common methods to access or manipulate HTML elements using the DOM.
- *JavaScript DOM Methods and Properties* (Matt, 2024), which provides examples of DOM methods and properties.



Thoughts on this page?

